

Learning Session Architecture Evolution v2

Purpose

Future development concept for the existing Learning module: add live learning sessions without making YouTube, Zoom, or Google Meet separate Learning domains.

Core Architecture

Learning remains the SaaS domain; external providers are session infrastructure.

Learning → Learning Session → Provider Adapter → YouTube / Zoom / Google Meet

Session Types

- BROADCAST — primarily YouTube Live.
- INTERACTIVE — primarily Zoom or Google Meet.
- ON_DEMAND — recorded learning content.

Providers

- YOUTUBE
- ZOOM
- GOOGLE_MEET
- INTERNAL — future option if the SaaS later owns live infrastructure.

Main Use Cases

YouTube Live: broadcast-style classes and larger audiences.

Zoom: interactive workshops/classes.

Google Meet: interactive classes, especially for Google-oriented users.

Learning Flow

Course/Curriculum → Learning Session → Scheduled/Live → Learner joins → Attendance/Progress → Assessment → Completion → Certificate.

Workspace Integration

Learning can become a Workspace training system with onboarding paths, course completion, live attendance, certification, and training dashboards. RBAC must follow the existing Workspace authorization model.

Candidate Entity

learning_sessions:

- id
- course_id / learning_id
- instructor_id
- workspace_id
- session_type
- provider
- external_session_id
- scheduled_at
- started_at

- ended_at
- status
- timestamps

Provider credentials/tokens require secure secret handling.

Candidate Status

DRAFT, SCHEDULED, LIVE, ENDED, CANCELLED, FAILED.

Activity Log

Candidate events include session created/updated/scheduled/started/ended/cancelled, provider connected/disconnected, attendance synchronization, and other important session changes.

Architecture Principles

1. SaaS is not the video-streaming server in the initial phase.
2. Use provider adapters.
3. Use caching/synchronization rather than aggressive polling.
4. Verify API, OAuth scopes, account plans, quotas/rate limits and policies during implementation.

Future Scope

Automated attendance, progress synchronization, certificate eligibility, live chat/Q&A;, AI Learning Assistant, recording-to-on-demand, multi-provider session management, and Workspace training analytics.

Required Impact Analysis

Assess existing Learning PRD, ERD/schema, RBAC/permissions, user flow, API contracts, Workspace, Activity Log, notifications, SEO, and billing/subscription if later monetized.

Note: This is future architecture context, not a final ERD, schema, API contract, permission matrix, or ADR.